

Germanium-rich nanoparticles in Cu-poor sphalerite: A new mechanism for Ge enrichment

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ABSTRACT

Germanium (Ge) is a critical raw material used in high-technology industry (i.e., optical industry) applications, and it is predominantly concentrated in coals and Zn-rich deposits. Previous studies on Zn-rich deposits have documented a correlation between Ge enrichment and the Cu, Ag, and/or Pb-Mn contents in the sphalerite crystal lattice. In this study, we observed Ge-rich nanoparticles hosted in Cu-poor sphalerite from the Banbianjie Zn-Ge deposit (>800 t graded at ~100 ppm Ge), located in southwest China. Laser-ablation-inductively coupled plasma-mass spectroscopy (LA-ICP-MS) analyses revealed that sphalerite contains very heterogeneous Ge contents (172–1553 ppm). Germanium contents showed positive correlations with Fe, Mn, and Pb contents and negative correlations with Cd contents. Higher Ge contents were detected in the darker zones, whereas the lighter zones showed systematically low Ge contents and were enriched in Cd. Using transmission electron microscopy (TEM), Zn-Ge-Pb-S nanoparticles were identified in the darker zones of sphalerite. These nanoparticles exhibited Ge/Pb ratios (0.48–1.96) very similar to those measured in sphalerite (0.36–2.04), suggesting that Ge could be essentially hosted within the nanoparticles. We propose that the amounts of Zn-Ge-Pb-S nanoparticles are related to a self-organization model induced by rapid crystal growth. This self-organization processes may control the fluctuations of element concentrations in the boundary

layer. This study highlights the importance of studying the nanoscale expression of critical elements to understand their incorporation mechanisms into natural materials.

INTRODUCTION

Critical raw materials such as germanium (Ge) are key factors in the high-tech industry with various applications in the so-called “green” technologies (Gulley et al., 2018; Wang, 2019; Zhai et al., 2019; Cugerone et al., 2020; Zhou et al., 2021b; Kelley et al., 2021), such as infrared goggles for military use (Gulley et al., 2018), high-performance fiber-optic cables for internet, and semiconductor materials for multijunction solar cells (Frenzel et al., 2014). Germanium is mainly extracted as a by-product of coal (Seredin et al., 2006; Du et al., 2009; Hu et al., 2009) and base metal (Pb-Zn) deposits, mostly from sphalerite (Frenzel et al., 2014, 2017; Paradis, 2015; Lawley et al., 2022; Torró et al., 2023). A significant increase in the production of Ge has occurred in the past 20 yr from 45 t/yr to 140 t/yr (U.S. Geological Survey, 1996, 2022). To deal with this increasing demand, the general distribution and mechanism(s) of concentrations of Ge in Zn ore need to be better understood to constrain the Ge resource potential of given base metal deposits in an economically and environmentally sustainable manner (Liu et al., 2023).

Carbonate-hosted base metal (Pb-Zn) deposits associated with low-temperature fluids (<150 °C) are generally richer in Ge (Fig. 1). For example, the world-class Huize carbonate-hosted deposit contains 525 t graded at 93 ppm Ge (Fig. 1; Tao et al., 2019). Sphalerite is the main host mineral for Ge in carbonate-hosted deposits. Previous studies have revealed that Ge is incorporated into the sphalerite crystal lattice through substitution of Zn²⁺ coupled with monovalent cations such

as Cu⁺ and Ag⁺ (Johan, 1988; Cook et al., 2009; Belissont et al., 2014, 2016; Bonnet et al., 2016; Bauer et al., 2019; Torró et al., 2023) and more complex substitutions involving divalent cations such as Pb²⁺ and/or Mn²⁺ (Luo et al., 2022). It is noteworthy that the metamorphism of Ge-rich sphalerite may induce the redistribution of Ge and the formation of micro- to nanoscale Ge minerals (e.g., brunogeierite and briartite; Höll et al., 2007; Cugerone et al., 2018, 2020; Fougereuse et al., 2023).

Newly reported Ge resources in the Huangsi region (SW China) reach >1200 t, mostly located in the Banbianjie (>800 t) and Zhulingou (>400 t) deposits (Fig. 1; Zhou et al., 2021a; Luo et al., 2022). More specifically, the Banbianjie deposit contains resources of 0.8 Mt graded at 1.78%–9.5% Zn and mean grades of ~100 ppm Ge (An et al., 2022). Previous studies constrained the origin and geological conditions of the Ge enrichment in sphalerite in the Banbianjie and Huangsi region (An et al., 2022; Luo et al., 2022) without reporting the micro- to nanoscale distribution of Ge in sphalerite.

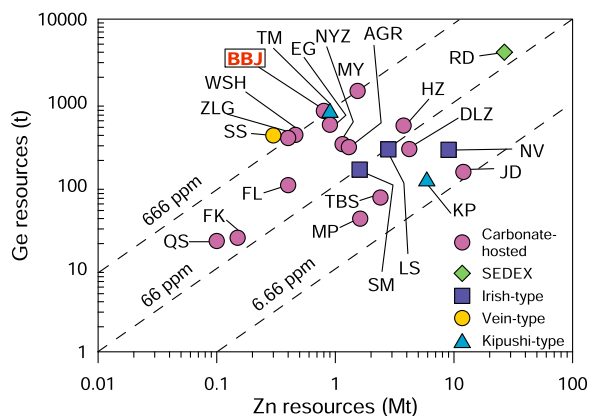
Here, we focused on the spatial distribution of Ge in sphalerite. Focused ion beam–scanning electron microscopy (FIB-SEM) and transmission electron microscopy (TEM) analyses were applied to characterize the nanoscale distribution of Ge in sphalerite. We also propose a schematic model for the enrichment of Ge in the Banbianjie deposit.

REGIONAL GEOLOGY

The southeast margin of the Yangtze block (Fig. 2A) contains late Proterozoic metamorphic and metasedimentary rocks and Cambrian to Permian–Triassic sedimentary rocks (Fig. 2B). The Proterozoic metamorphic rocks belong to the Banxi Group and include graywacke, slate, and

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Mayuan, China; NV—Naven, Ireland; NYZ—Nayongzhi, China; QS—Qingshan, China; RD—Red Dog, Alaska, USA; SEDEX—sedimentary exhalative; SM—Silvermines, Ireland; SS—Saint-Salvy, France; TBS—Tianbaoshan, China; TM—Tsumeb, Namibia; WSH—Wushihe, China; ZLG—Zhulingou, China.

schist. The Banxi Group has undergone low-grade greenschist-facies metamorphism (Wang et al., 2010). U-Pb zircon ages of tuff and tuffaceous rocks in the Banxi Group constrain its deposition between ca. 820 Ma and 715 Ma (Lan et al., 2014,

2015; Li et al., 2019; Xian et al., 2020). The Cambrian to Silurian sedimentary succession consists of a shale-carbonate-siliciclastic transition, indicating either a regression of sea level or the uplift of the southeastern Yangtze block (Yao and Li,

2016). The Devonian to Triassic succession comprises sandstone, wackestone, dolostone, limestone, and shale and records at least three cycles of transgression-regression (GBGMR, 1987).

The structural framework of the Huangsi region is dominant by N-S-, E-W-, and NE-SW-trending structures (Fig. 2B). The N-S-striking structures dominantly comprise box anticlines (e.g., the Huangsi and Wangsi anticlines) and chevron synclines (e.g., the Duyun syncline; Fig. 2B). These structures are frequently cut by N-S-striking thrust faults (Fig. 2B). The Wangsi anticline is cut by N-S- and NE-SW-striking normal faults, whereas the Huangsi anticline is cut by E-W-striking normal faults (e.g., Huangsi fault; Fig. 2B).

BANBIANJIE DEPOSIT, SOUTHWEST CHINA

Deposit Geology

The Banbianjie district is mainly composed of Silurian to Middle Devonian clastic rocks, Upper Devonian dolostone, Carboniferous car-

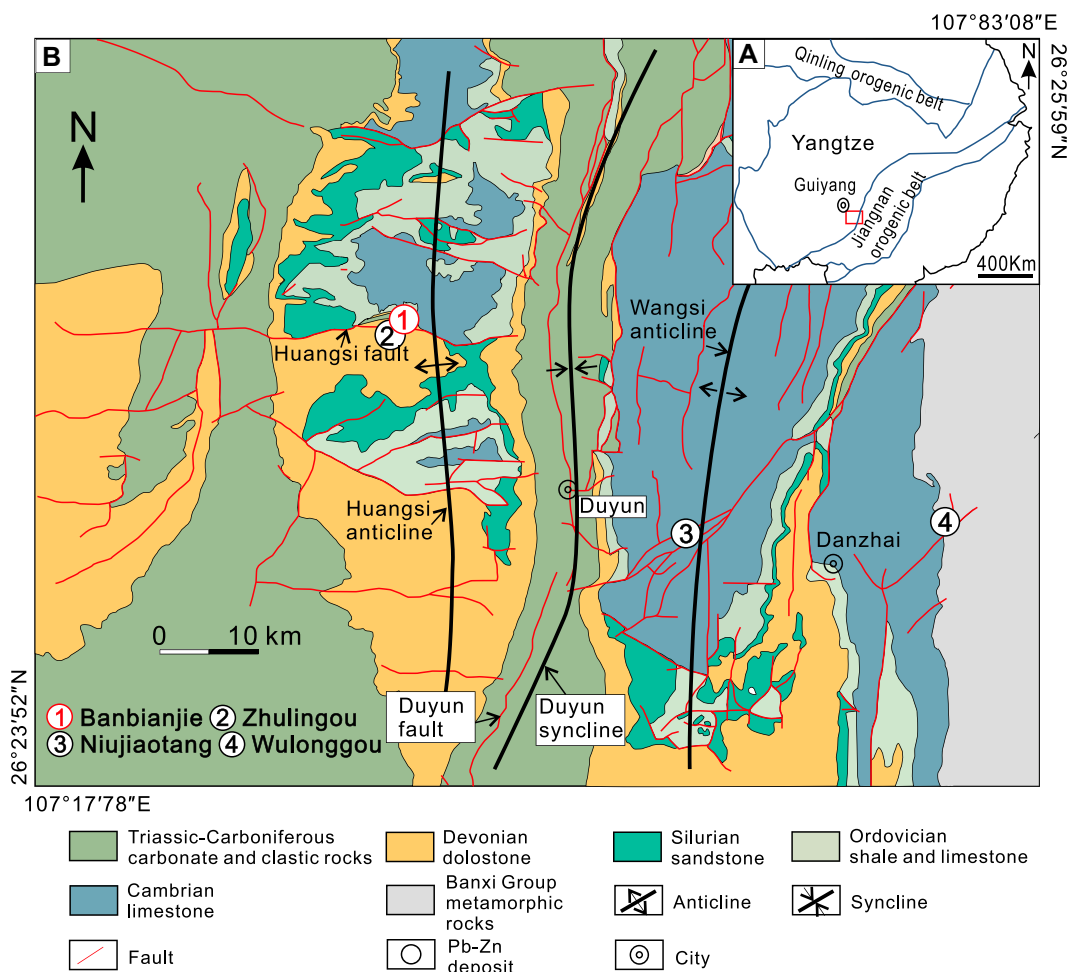


Figure 2. (A) Structural map of the southeastern margin of the Yangtze block, South China. Red rectangle represents the study area. (B) Geological map of the SE Yangtze block with the location of the Banbianjie deposit.

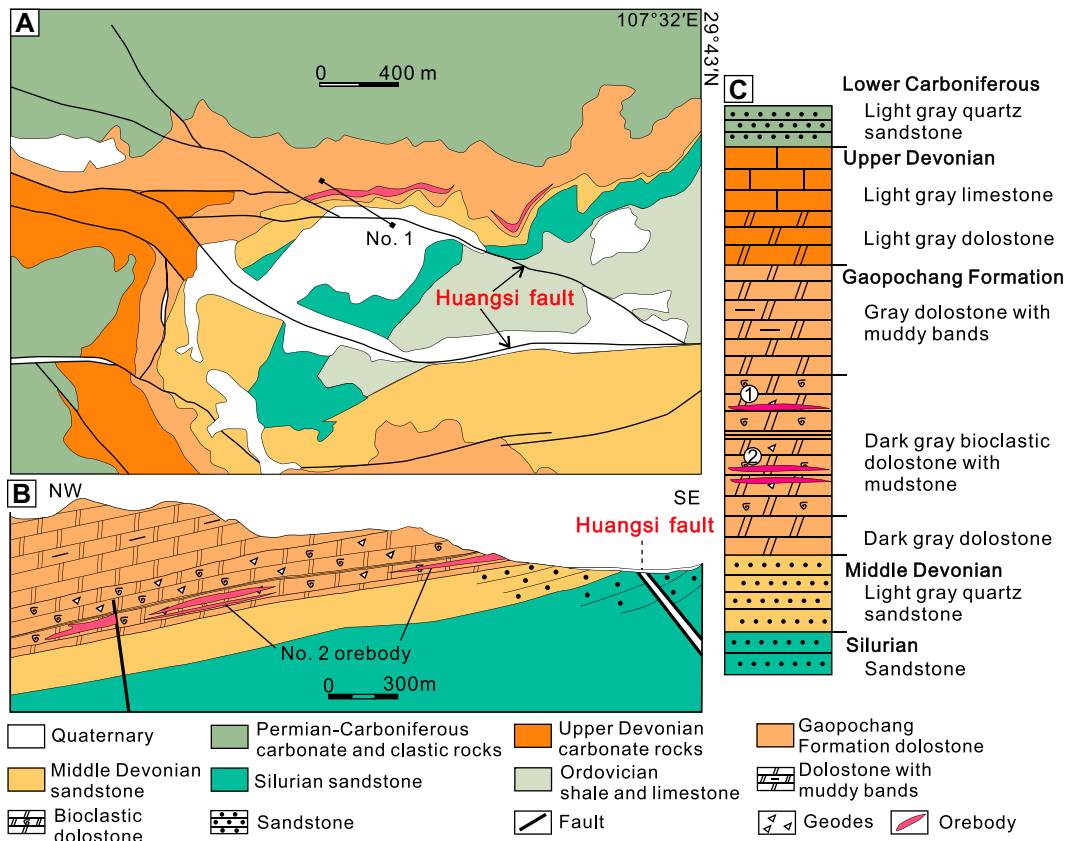


Figure 3. (A) Geological map of the Banbianjie deposit. (B) Cross section of No. 1 exploration line. (C) General stratigraphy of the Banbianjie area.

bonate, and Permian clastic rocks (Chen et al., 2006). The ore bodies are hosted in the Devonian Gaopochang Formation (Fig. 3A), which shows a variable thickness in the range between 380 m and 1074 m. The biostratigraphy suggests that the Gaopochang Formation was deposited between ca. 385 Ma and 360 Ma (Chen et al., 2006). The lower part of the Gaopochang Formation consists of dark-gray, fine-grained dolostone and argillaceous dolostone (Figs. 3B and 3C). The middle part is characterized by dark-gray, thick-bedded, bioclastic dolostone interbedded with restricted black, grayish-yellow, and grayish-green mudstone (Figs. 3B and 3C). A characteristic feature of the dolostone in the middle part is the abundance of centimeter-scale geodes composed of dolomite (Figs. 4A and 4B). The upper part of the Gaopochang Formation is dominantly composed of thick, light-gray or gray, fine-grained dolostone, with a minor proportion of argillaceous dolostone. The Zn-Ge mineralization occurs mainly in the middle part of the Gaopochang Formation (Figs. 3B and 3C).

In the mining area, the main structures are the E-W-striking Banbianjie anticline and the Huangsi fault zone. The Huangsi fault zone crosscuts the south limb of the Banbianjie anticline (Fig. 3A; Chen et al., 2006) and comprises a set of E-W-striking faults with a general dip

of 40°–70° to the south. Chen et al. (2006) suggested that activity on the fault zone was syndimentary and controlled the lithofacies of the Gaopochang Formation. However, the fact that the Carboniferous–Triassic rocks are crosscut by the Huangsi fault would indicate that the Huangsi fault was reactivated after the Carboniferous.

Ore Bodies

The Zn-Ge mineralization, which is hosted exclusively in the middle part of the Gaopochang dolostone, comprises six stratabound and lenticular ore bodies. Mineralization is hosted in bedded dark-gray dolostone with a fine-crystalline texture (Figs. 4A and 4B). The No. 2 ore body, the largest one in Banbianjie, has a thickness ranging from 1.13 m to 11.8 m, and it contains ore with Zn grades between 1.78% and 9.5% and Ge grades of ~100 g/t. The ore bodies are composed of multiple Zn-rich veins that are parallel to or crosscut the bedding of the dark-gray dolostone (Figs. 4A and 4B). These veins vary in thickness, from 1 cm to 50 cm, and in texture, locally showing brecciated texture (Fig. 4C). In the breccia, the clasts are rounded and cemented by sulfide minerals. Sulfides in the veins show macroscopic oscillatory zoning (Fig. 4D).

SAMPLING AND ANALYTICAL METHODS

Samples from the Banbianjie deposit were collected from underground galleries and drill cores. Forty-five thin sections were studied with optical microscopy and scanning electron microscopy (SEM). Among them, sphalerite in samples BBJ-1, BBJ-5, and ZK1008 was analyzed by laser-ablation-inductively coupled plasma-mass spectroscopy (LA-ICP-MS) for trace elements. Colloform sphalerite in samples BBJ-1 and ZK1008 was selected for elemental mapping by LA-ICP-MS.

SEM Analysis

SEM analyses were carried out on a JSM7800F SEM at the State Key Laboratory of Ore Deposit Geochemistry, Institute of Geochemistry, Chinese Academy of Sciences (IGCAS), Guiyang, China. The backscattered electron (BSE) images were collected with a voltage of 20 kV.

FIB/SEM and TEM Analyses

The FIB combined with SEM imaging was conducted on an FEI Scios dual-beam FIB/SEM

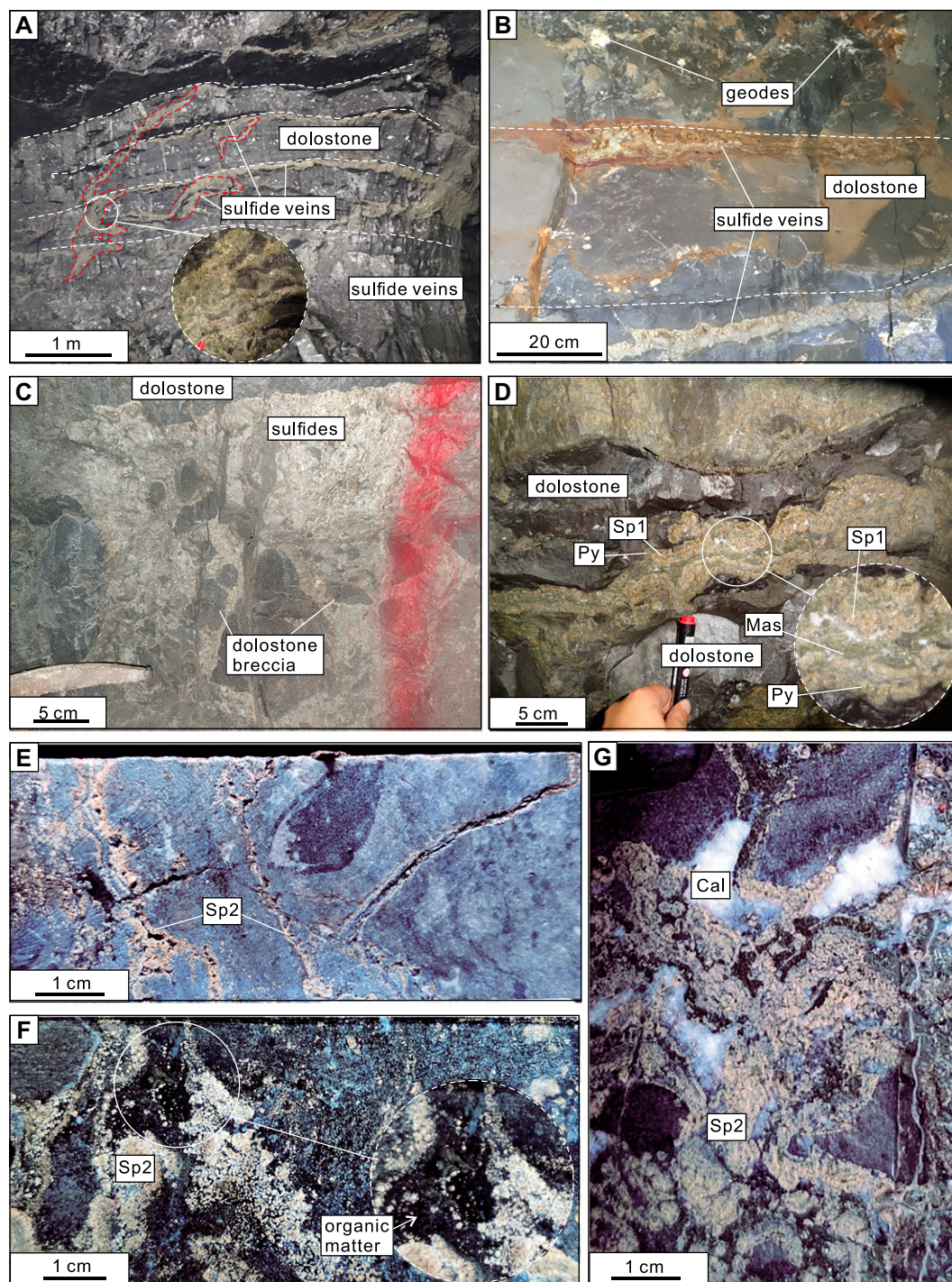


Figure 4. Field photographs of the Zn-Ge mineralization in the Banbianjie deposit. (A) Stratobound sulfide with local veins crosscutting dolostone bedding (white dotted lines). Red dotted lines highlight the sulfide veins crosscutting bedding. (B) Stratabound sulfide veins parallel to bedding. (C) Stratobound sulfide veins with few dolostone brecciated clasts with rounded shapes. The breccia is cemented by a sulfide mass. (D) Colloform sphalerite, pyrite, and marcasite bands. Dolomite filled the empty space. (E) Sp2 veinlets filling fractures in dolostone. (F) Lumpy aggregates of Sp2 with a significant presence of organic matter. (G) Sp2-calcite veins. Cal—calcite; Mas—marcasite; Py—pyrite; Sp1—sphalerite 1; Sp2—sphalerite 2.

at the IGCAS. The FIB/SEM system was used for the preparation of TEM foils. A sphalerite area with ~400 ppm of Ge was selected for the FIB milling. The TEM foil had a dimension of $10 \times 10 \times 0.06 \mu\text{m}$.

The TEM analysis was conducted on an FEI Tecnai G2 F20 TEM equipped with a Gatan imaging filter, an energy-dispersive X-ray analyzer with an ultrathin window, and a Fischione

high-angle annular dark-field (HAADF) detector at the IGCAS. The HAADF images were taken by a camera with a length of 75 mm, which can avoid the influence of Bragg scattered electrons. Energy-dispersive X-ray spectroscopy (EDS) was conducted using the TEM in scanning transmission mode. The acquisition time was 120–240 s. The detector conditions included an energy resolution of 134 eV, a reference energy

of 5.9 keV, a detector distance of 11.8 mm, and a detector angle of 14.6° . The microscope conditions included an acceleration voltage of 200 kV and a beam current of 1 nA. The spot sizes were 5–10 nm to avoid the influence of the host. The spectral lines analyzed for each element were S ($K\alpha$), Fe ($K\alpha$), Zn ($K\alpha$), Ge ($K\alpha$), and Pb ($L\alpha$). Further details of the analytical procedure can be found in Liang et al. (2021).

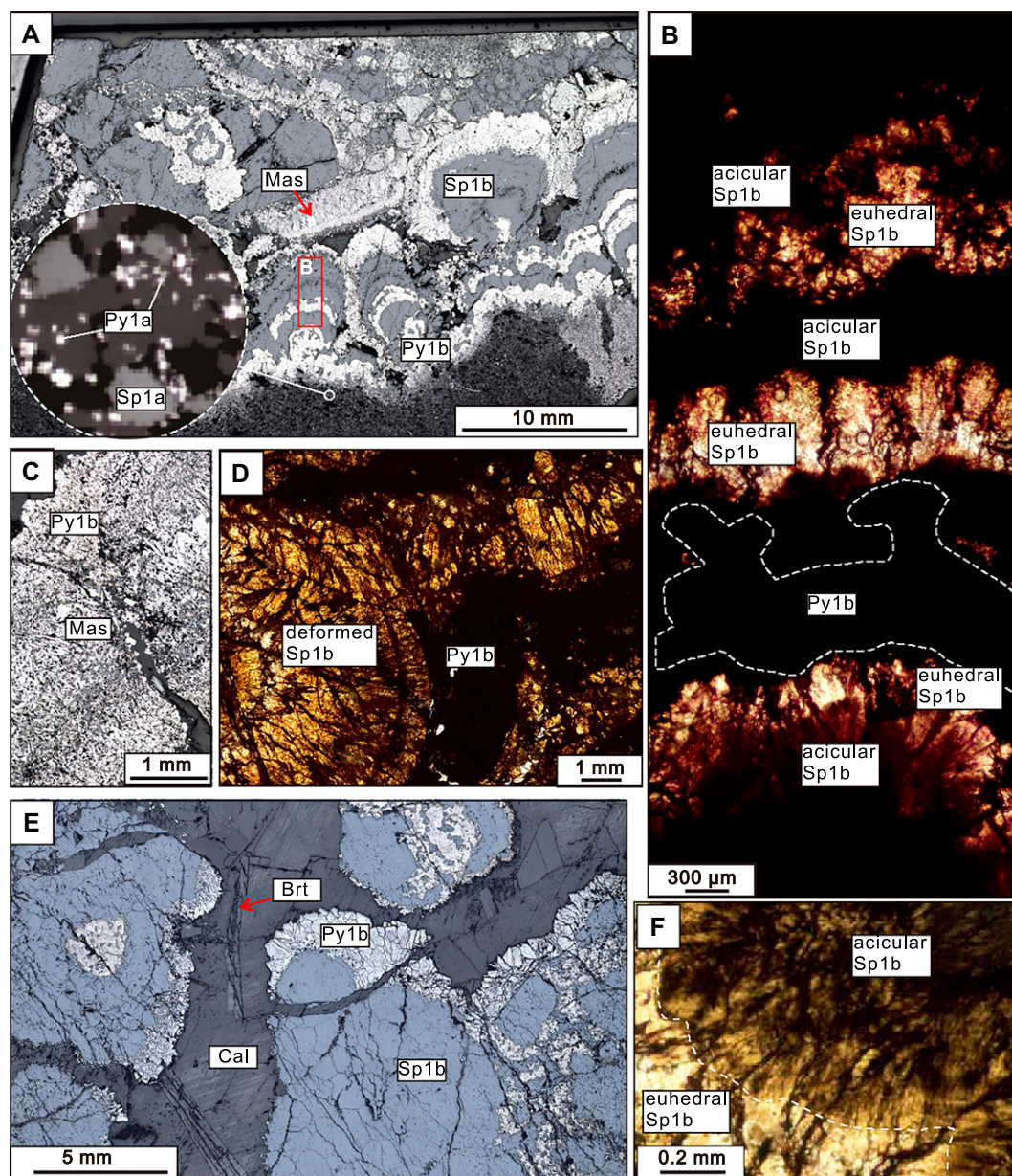


Figure 5. Representative photomicrographs showing textures of type 1 ore. (A) Disseminated sphalerite (Sp1a) and pyrite (Py1a) in host rocks, followed by colloform sphalerite (Sp1b) and pyrite (Py1b) assemblages. (B) Oscillatory zoned Sp1b in transmitted plane-polarized light. Dark Sp1b is acicular, whereas the light Sp1b is mostly euhedral. (C) Euhedral Py1b intergrown with acicular marcasite. (D) Deformed Sp1b with intense microfractures. (E) Colloform Sp1b overgrown by Py1b, followed by calcite and euhedral barite. (F) Oscillatory zoned Sp1b with darker color and acicular texture in the core, and lighter color and euhedral texture on the rim. Brt—barite; Cal—calcite; Mas—marcasite; Py1a—pyrite 1a; Py1b—pyrite 1b; Sp1a—sphalerite 1a; Sp1b—sphalerite 1b.

LA-ICP-MS Trace-Element Analyses

Trace-element contents of sphalerite were determined by LA-ICP-MS at the Guangzhou Tuoyan Analytical Technology Co., Ltd., Guangzhou, China. The samples were analyzed using an iCAP RQ ICP-MS instrument equipped with a NWR ablation system ($\lambda = 193$ nm). The operating conditions were a frequency of 8 Hz and a laser spot size of 30 μm . NIST610, GSE-2G, and MASS-1 were used as external standards for calibration, and ^{66}Zn ($\text{Zn} = 650,000$ ppm) was used as the internal standard. Each analysis consisted of 50 s of background measurement and 40 s of sample ablation (signal measurement). The time-inte-

grated LA-ICP-MS signals were processed using Iolite v3.25 (Paton et al., 2011).

For LA-ICP-MS mapping, the operating conditions were a fluence of 1.5 J/cm², a repetition rate of 40 Hz, and a washout time of 20 s. Two laser-spot sizes were used depending on the thickness of the zoning bands (15 μm spot size measurements are used for sphalerite 1b [Sp1b]; 20 μm spot size measurements are used for sphalerite 2 [Sp2]). To produce the elemental maps, adjacent lines were ablated successively. A square-shaped laser aperture was progressively translated by a continuous movement of the stage under the fixed ablation site to generate an evenly ablated line (see Zhou et al., 2017). NIST 610, GSE-2G, and

MASS-1 were analyzed as external standards (Wilson et al., 2002). The external standards were ablated three times, before, in between, and after the analysis mapping of each area. Each analytical procedure involved the analysis of 16 elements: ^{34}S , ^{55}Mn , ^{57}Fe , ^{59}Co , ^{60}Ni , ^{65}Cu , ^{66}Zn , ^{71}Ga , ^{74}Ge , ^{75}As , ^{77}Se , ^{107}Ag , ^{111}Cd , ^{115}In , ^{118}Sn , and ^{208}Pb . The elemental maps were generated using Iolite v3.25, which runs with Igopro (Paton et al., 2011). Further details on the analytical procedure can be found in Zhou et al. (2017).

MATLAB-based package XMapTools was applied to quantify and extract the measured contents from the elemental mapping data. The quantified data for each element (ppm) were first

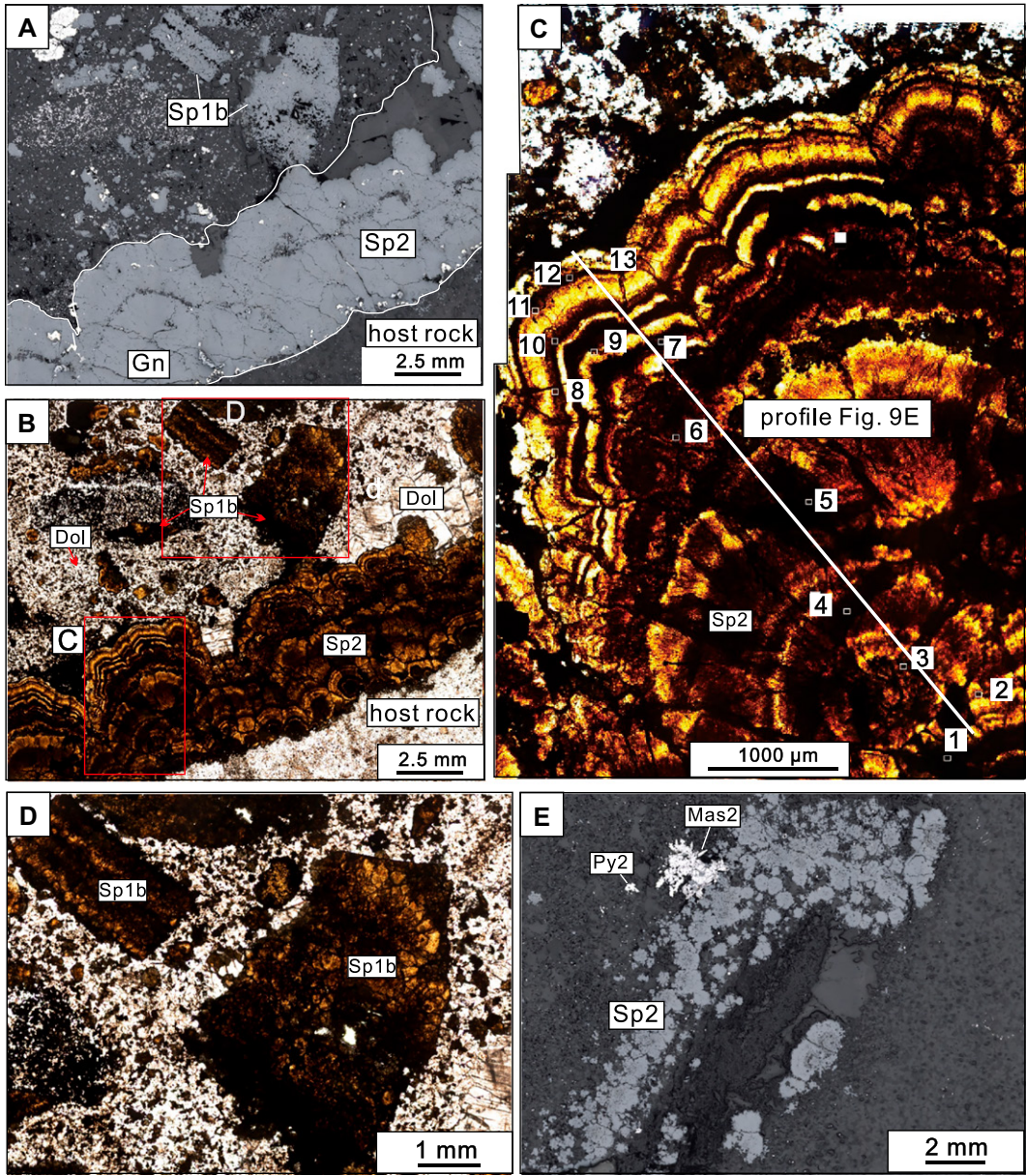


Figure 6. Representative photomicrographs showing the texture of type 2 ore. (A) Sp2-dolomite vein and brecciated fragments of type 1 ore (reflected light). Sp1b and Sp1b-Py1b brecciated clasts have an angular shape. (B) Sp2-dolomite veins and brecciated fragments of Sp1b (transmitted light). Type 1 ore clasts are cemented by dolomite. (C) Oscillatory zoning of Sp2 with the location of the laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) trace-element data points. (D) Brecciated fragments of Sp1b cemented by dolomite. (E) Colloform Sp2 along fractures with minor Py2 and Mas2. Dol—dolomite; Gn—galena; Mas2—marcasite 2; Py2—pyrite 2; Sp1b—sphalerite 1b; Sp2—sphalerite 2.

Minerals	Stage 1	Stage 2
Sphalerite	— Sp1a acicular-euhedral Sp1b	— acicular Sp2
Pyrite	—	—
Marcasite	—	—
Galena	—	—
Dolomite	—	—
Barite	—	—
Calcite	—	—

Figure 7. Paragenetic sequence of the Banbianjie deposit. Sp1a—sphalerite 1a; Sp1b—sphalerite 1b; Sp2—sphalerite 2.

exported using Iolite. XMapTools can visualize and explore these data via line model sampling functions, via pixel identification maps, or via histogram, binary, and/or ternary diagrams (see Lanari et al., 2014; Raimondo et al., 2017; Sun et al., 2021).

RESULTS

Texture and Mineralogy

Two types of ore were recognized in the Banbianjie deposit (Figs. 4A–4G). Type 1 ore consisted mostly of centimeter-scale veins (Figs. 4A, 4B, and 4D) with red to brown colloform sphalerite (Sp1b; Figs. 4D

TABLE 1. SUMMARY OF TRACE ELEMENTS IN DIFFERENT TYPES OF SPHALERITE FROM THE BANBIANJIE DEPOSIT, SOUTHWEST CHINA

Sphalerite type	Mn (ppm)	Fe (ppm)	Cu (ppm)	Ga (ppm)	Ge (ppm)	As (ppm)	Ag (ppm)	Cd (ppm)	In (ppm)	Sn (ppm)	Hg (ppm)	Tl (ppm)	Pb (ppm)	Ge/Pb (mole ratio)
Sp1a														
Max.	41.3	9774	1.89	3.43	683	0.48	2.36	485	0.26	7.86	22.3	36.6	1287	1.65
Min.	34.7	7799	1.67	1.66	579	0.24	1.89	472	0.001	2.05	18.6	28.4	986	1.42
Median	40.7	8333	1.82	1.97	649	0.32	2.11	476	0.004	2.66	21.5	33.1	1199	1.60
Mean	38.9	8635	1.79	2.35	637	0.35	2.12	478	0.09	4.19	20.8	32.7	1157	1.56
Q1	37.7	8066	1.74	1.82	614	0.28	2.00	474	0.002	2.35	20.0	30.8	1092	1.51
Q3	41.0	9053	1.85	2.70	666	0.40	2.23	481	0.13	5.26	21.9	34.8	1243	1.63
STD	3.65	1022	0.11	0.95	53.2	0.12	0.23	6.57	0.15	3.20	1.96	4.09	155	0.12
Sp1b														
Max.	133	33,447	2.86	47.7	1,553	6.69	5.55	2,670	8.02	23.6	32.4	102	6579	2.48
Min.	7.74	2131	0.29	2.27	174	0.07	0.30	59	b.d.l.	0.11	2.11	7.16	244	0.47
Median	49.7	8581	0.55	7.70	736	0.55	2.21	508	0.85	6.52	16.8	31.9	1429	1.04
Mean	52.7	9281	0.75	11.6	720	0.92	2.24	613	1.52	7.76	17.7	43.0	2128	1.20
Q1	34.6	7050	0.40	5.57	446	0.20	1.63	399	0.17	3.39	12.5	21.2	1075	0.76
Q3	71.1	11,014	0.84	12.8	954	0.93	2.87	641	2.25	11.9	26.1	67.4	3207	1.54
STD	28.1	5606	0.54	11.3	347	1.38	1.16	532	1.90	6.24	8.35	28.1	1721	0.53
Sp2														
Max.	79.1	22,245	19.2	21.2	1,150	5.30	0.52	2,863	0.88	9.22	3.38	56.1	5949	1.69
Min.	21.0	1857	0.41	1.95	306	0.17	0.28	232	0.01	0.41	1.46	5.41	1100	0.48
Median	41.2	11,224	0.70	10.9	666	1.23	0.37	651	0.05	4.86	2.53	23.6	1683	1.13
Mean	41.7	11,777	2.01	11.6	706	1.76	0.38	905	0.11	4.70	2.52	25.5	1970	1.10
Q1	35.1	9279	0.58	7.74	553	0.96	0.35	479	0.02	3.37	2.29	20.0	1471	0.88
Q3	44.3	13,072	0.99	14.8	837	1.73	0.40	1,104	0.07	6.25	2.70	29.6	2002	1.29
STD	11.9	4170	4.19	5.38	214	1.42	0.07	681	0.20	2.42	0.41	10.48	1003	0.32

Note: b.d.l.—below detection limit; Max.—maximum; Min.—minimum; Q1—quartile 1; Q3—quartile 3; STD—standard deviation.

and 5A–5E), pyrite (Py1b), and acicular marcasite with minor dolomite, barite, and calcite. The thicknesses of individual sphalerite and pyrite layers varied from 1 mm to 5 mm (Figs. 5A and 5E). Under the microscope with transmitted light, the colloform Sp1b showed oscillatory color zoning with black to light-yellow color zones. The black-to-brown-colored Sp1b crystals were acicular and oriented perpendicular to the color zoning boundaries (Figs. 5B and 5F). However, light-yellow-colored Sp1b generally consisted of euhedral sphalerite crystals (Figs. 5B and 5F). Acicular marcasite crystals were intergrown with euhedral pyrite aggregates (Fig. 5C). Locally, some sulfide-rich areas were more fractured,

and deformed textures were observed with characteristic coarse sphalerite surrounded by fine sphalerite crystals (<100 μm ; Fig. 5D). Type 1 ore was locally composed of subhedral sphalerite and pyrite crystals disseminated in the dolostone host (Sp1a and Py1a; Fig. 5A).

Type 2 ore occurred as veinlets filling fractures or as globular aggregates generally thinner than type 1 ore (Figs. 4E–4G); it was composed of colloform Sp2, with minor pyrite (Py2), marcasite (Mas2), and galena in a gangue of dolomite and calcite (Figs. 6A and 6E). Angular sphalerite clasts of type 1 ore were cemented by dolomite and crosscut by the type 2 ore (Figs. 6A–6D). Sp2 comprised oscillatory zones of yellow to brown colors (Figs. 4E–4G and 6C). In general,

oscillatory color zoning textures in acicular Sp2 were more complex and intense compared to Sp1 (Figs. 6B, 6C, and 7).

Chemical Composition of Sphalerite

Spot Analyses

LA-ICP-MS results of sphalerite included three analyses of Sp1a, 28 analyses of Sp1b, and 23 analyses of Sp2 (Table 1; Supplemental Material¹).

Sp1a contained relatively high Fe, Cd, Ge, and Pb contents with 7798–9773 ppm, 471–484 ppm, 578–683 ppm, and 985–1286 ppm, respectively (Supplemental Material). Detectable Mn and Tl contents are also observed in Sp1a (34–41 ppm and 28–36 ppm, respectively). The contents of Cu, Ga, Ag, and Sn were systematically below 10 ppm (1.67–1.89 ppm, 1.66–3.43 ppm, 1.89–2.35 ppm, and 2.04–7.86 ppm, respectively). Germanium contents were higher in Sp1b than in Sp1a and ranged between 236 ppm and 1553 ppm, with a mean value of 720 ppm. Contents of Mn, Fe, Ag, Cd, and Tl in Sp1b were similar to those of Sp1a (Fig. 8), whereas the Ga, Ge, In, and Pb contents were higher than those of Sp1a (Fig. 8). Compared with Sp1a and Sp1b, Sp2 exhibited similar contents of Mn (21.0–79.1 ppm), Fe (7716–19,466 ppm), Cu (0.41–19.2 ppm), Tl (5.41–56.1 ppm), and Pb (27.0–849 ppm), but higher contents

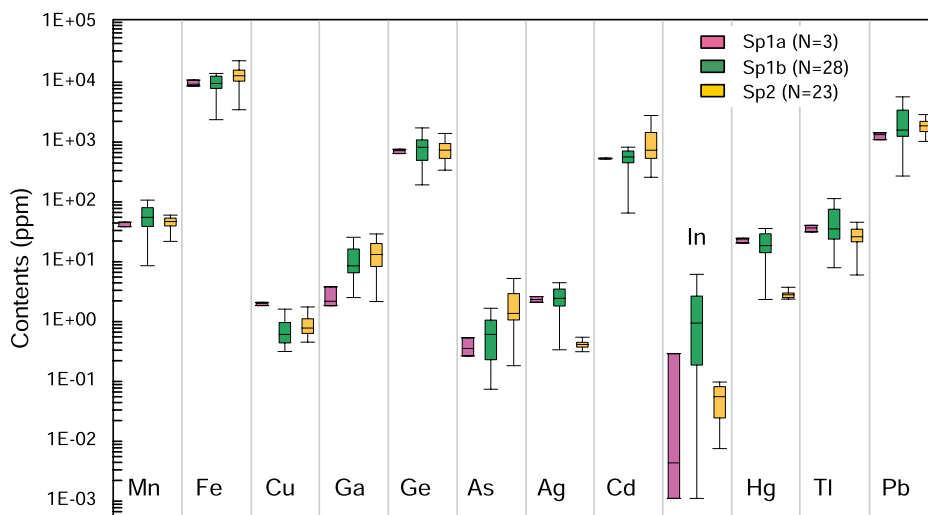


Figure 8. Box plot showing selected trace-element variations of sphalerite 1a (Sp1a), sphalerite 1b (Sp1b), and sphalerite 2 (Sp2).

¹Supplemental Material. LA-ICP-MS trace-element data of sphalerite from the Banbianjie deposit. Please visit <https://doi.org/10.1130/GSAB.S24291313> to access the supplemental material, and contact editing@geosociety.org with any questions.

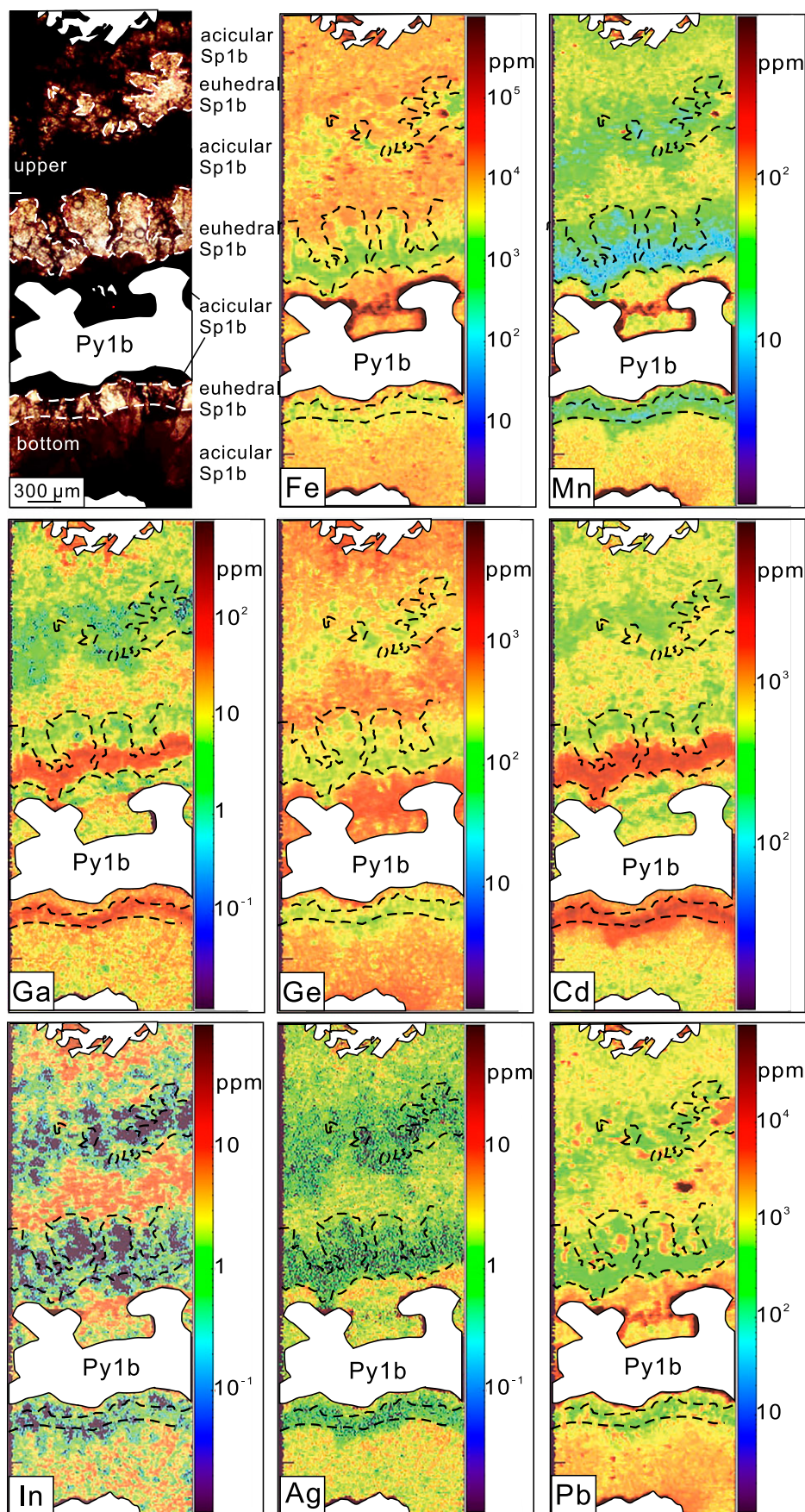


Figure 9. Laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) element maps of colloform sphalerite (Sp1b) and pyrite (Py1b).

of Ga (1.95–21.2 ppm), As (0.17–5.30), and Cd (232–2862 ppm), and lower contents of Ag (0.28–0.52 ppm) and Hg (1.46–3.38 ppm; Fig. 8). In Sp2, Ge contents ranged 306–1150 ppm, with a mean value of 706 ppm, which is similar to Sp1b.

Spatial Distribution of Elements

The spatial distributions of Fe, Mn, Ga, Ge, Cd, In, Ag, and Pb are illustrated for Sp1b and Sp2 in Figures 9, 10, and 11. The compositional maps revealed a very heterogeneous zoning in Sp1b and Sp2, frequently correlating with microscopic color zoning (Figs. 9 and 10). In Sp1b, the darker color zones were richer in Fe, Ge, Mn, In, Ag, and Pb and generally depleted in Cd and Ga; in contrast, the lighter color zones were frequently richer in Cd and Ga and depleted in Fe, Ge, Mn, In, Ag, and Pb compared to the dark color zones (Fig. 9). Nevertheless, local areas in the light color zones were also depleted in Cd and Ga. Compositional analyses performed along a transect (Figs. 11A–11D) showed that Ge contents were mainly correlated to Fe, Mn, and Pb and anticorrelated to Cd and Ga.

In Sp2, a more complex zoning was identified. The darker zones were enriched in Fe, Mn, Ga, Ge, In, and Pb, whereas the lighter zones were frequently enriched in Cd. The main difference in the trace-element distributions between Sp1b and Sp2 was that Ga was positively correlated with Cd in Sp1b (Figs. 11C and 11D) but was anticorrelated with Cd in Sp2 (Fig. 11E). More precisely, two types of zoning were observed in Sp2. In the bottom part, the zones with higher Ge contents had thicknesses from 500 μm to 2000 μm , i.e., thicker than the zones (300–600 μm) with lower Ge contents. In the upper part, the Ge-richer and Ge-poorer zones had similar thicknesses from 150 μm to 200 μm .

Compositions of Nanoscale Particles

At the nanoscale, the HAADF-TEM images performed on Sp1b revealed the occurrence of anhedral nanoparticles with sizes from 10 nm to 70 nm (Figs. 12A–12F). These nanoparticles were analyzed with EDS and consisted mainly of S, Zn, Ge, and Pb (Table 2). The contents of S, Zn, Ge, and Pb had ranges of 24.3–33.2 wt%, 32.3–58.9 wt%, 3.34–11.2 wt%, and 12.4–28.6 wt%, respectively.

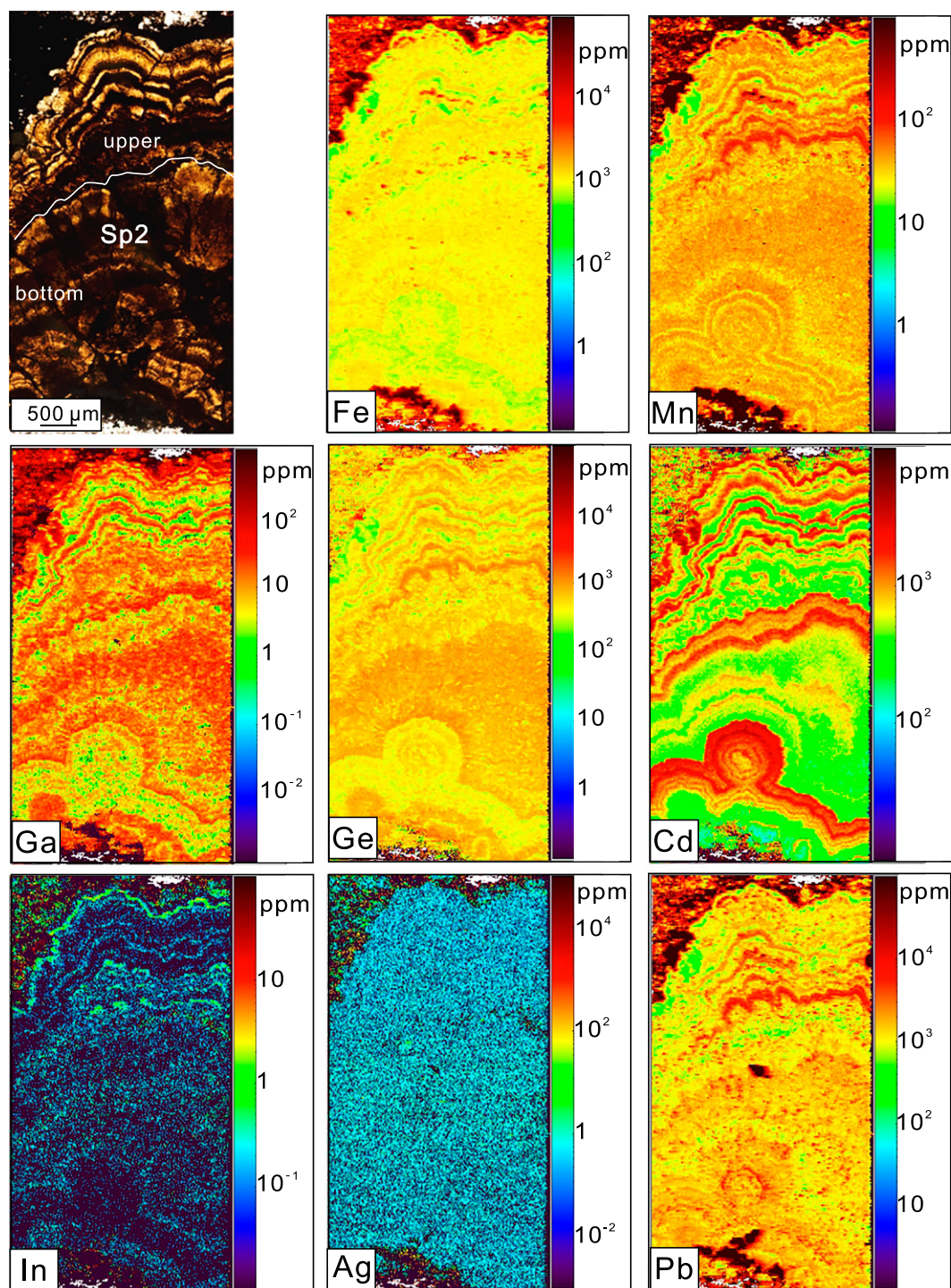


Figure 10. Laser ablation–inductively coupled plasma–mass spectrometry (LA-ICP-MS) elemental maps of sphalerite 2 (Sp2).

DISCUSSION

Germanium Incorporation in Sphalerite

In Cu–Ag–depleted Sp1 and Sp2 (Cu <19.2 ppm, Ag <5.6 ppm), Ge contents were positively correlated with Fe, Mn, and Pb (Figs. 13A, 13C, and 13E). This differs from the previously documented positive correlation

between Ge and Cu or Ag (Ye et al., 2011; Belisont et al., 2014; Bauer et al., 2019; Torró et al., 2023). At the microscale, the weak correlation ($R^2 = 0.14$) between Ge and Fe (Fig. 13A) in our analyses may exclude a $\text{Ge}^{4+} + \text{Fe}^{2+} + \square$ (vacancy) $\leftrightarrow 3\text{Zn}^{2+}$ coupled substitution (Cook et al., 2009; Luo et al., 2022). However, Ge was also positively correlated ($R^2 = 0.75$ and 0.56 , respectively) with Mn and Pb (Figs. 13C and

13E; 1:10 ratio for Mn), which may imply complex substitution mechanisms if Ge was hosted in the sphalerite lattice. Our study was therefore conducted with a nanoscale survey to determine the precise Ge host and incorporation mechanisms.

TEM analysis revealed the occurrence of Zn-, Ge-, Pb-, and S-rich nanoparticles within Sp1b. Unfortunately, Ge and Pb contents in sphalerite were not measured due to the high lower limits of

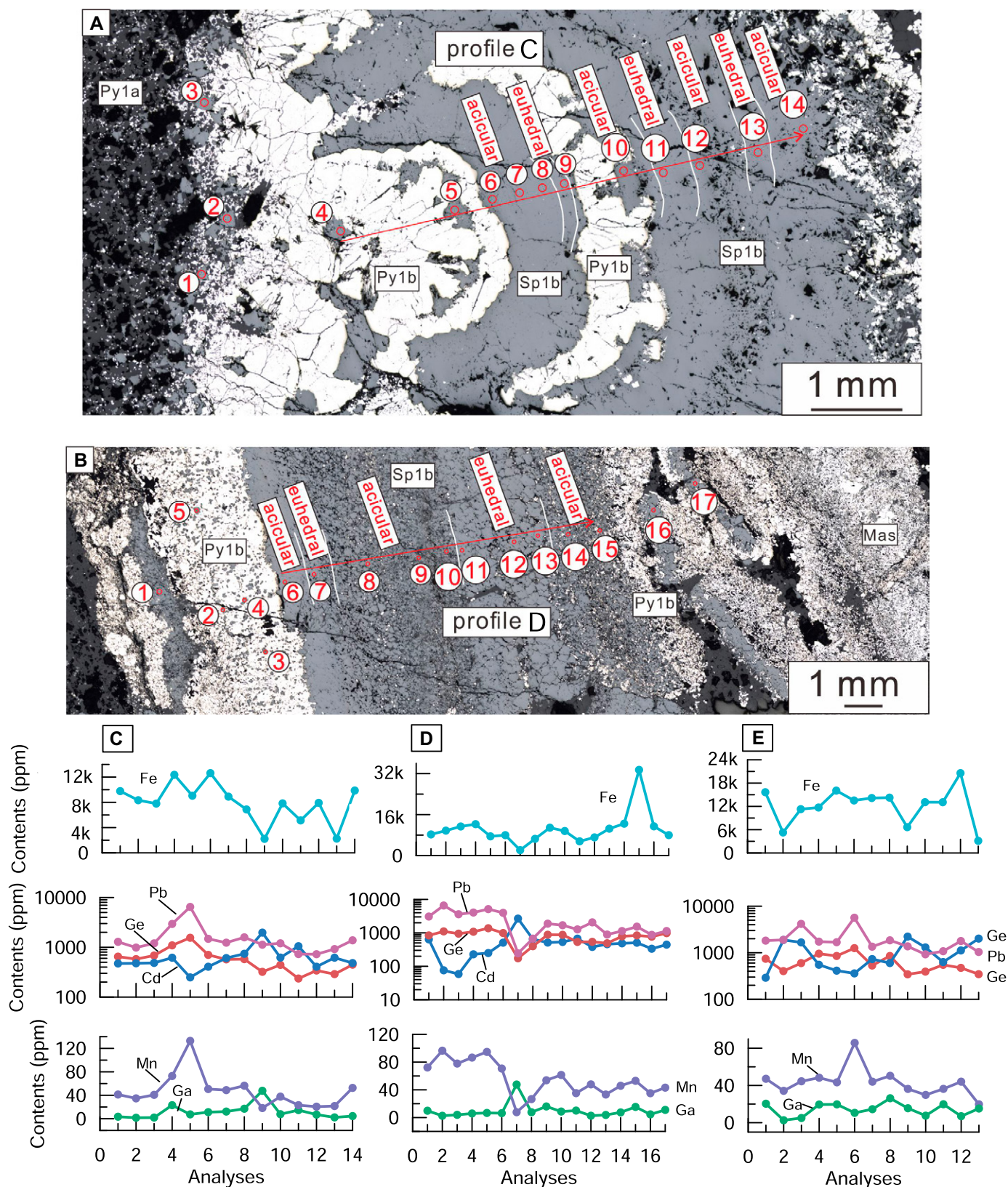


Figure 11. Transect showing trace-element variations in Sp1b and Sp2 from the Banbianjie deposit. (A and B) Location of laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) points in Sp1b. (C and D) Trace-element variations in Sp1b from the profile in panels A and B. (E) Trace-element variations in Sp2 (location of this profile is shown in Fig. 6C). Mas—marcasite; Py1a—pyrite 1a; Py1b—pyrite 1b; Sp1a—sphalerite 1a; Sp1b—sphalerite 1b; Sp2—sphalerite 2.

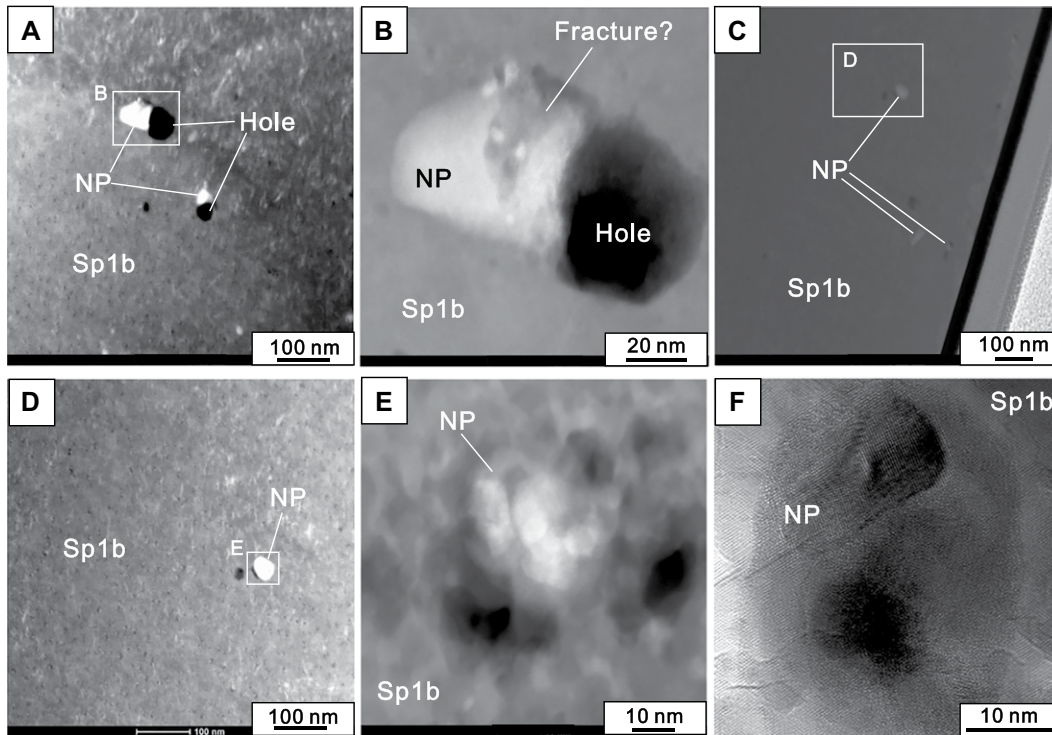


Figure 12. Transmission electron microscopy (TEM) images of Ge-rich nanoparticles in sphalerite 1b (Sp1b). (A) High-angle annular dark-field (HAADF) image showing two Ge-rich nanoparticles (NP) in Sp1b. (B) Detailed HAADF image showing a rounded Ge-rich NP. Elongated structure is interpreted as a fracture. (C) HAADF image showing three Ge-rich nanoparticles in Sp1b. (D) Close-up of image in panel C, showing that the nanoparticle is lighter than Sp1b in the HAADF image. (E) Close-up of image in panel D, showing that the Ge-rich nanoparticle seems to be an assemblage. (F) High-resolution transmission electron microscopy (HRTEM) image showing the planes of Ge-rich nanoparticle and Sp1b.

TABLE 2. TRANSMISSION ELECTRON MICROSCOPY-ENERGY-DISPERSIVE X-RAY SPECTROSCOPY (TEM-EDS) DATA OF Ge-RICH NANOPARTICLES HOSTED IN Sp1b FROM THE BANBIANJIJE DEPOSIT

Spot no.	Element	Weight (%)	Atomic (%)	Uncertainty (%)	Formula	Possible explanation	Ge/Pb (mole ratio)
6-1	S(K)	29.65	54.91	1.17	$Zn_{0.5}Ge_{0.16}Pb_{0.15}S$	$3ZnS \cdot GeS_2 \cdot PbS$	1.12
	Zn(K)	30.53	27.72	1.00			
	Ge(K)	11.20	9.16	0.72			
	Pb(L)	28.61	8.19	1.38			
6-2	S(K)	33.23	54.26	0.97	$Zn_{0.7}Ge_{0.08}Pb_{0.06}Ca_{0.01}S$	$9ZnS \cdot GeS_2 \cdot PbS$	1.25
	Ca(K)	0.41	0.54	0.13			
	Zn(K)	46.76	37.44	1.13			
	Ge(K)	5.96	4.30	0.47			
	Pb(L)	13.62	3.44	0.91			
7-1	S(K)	23.58	41.6	0.78	$Zn_{1.2}Ge_{0.1}Pb_{0.08}S$		1.14
	Fe(K)	0.22	0.22	0.12			
	Zn(K)	58.9	50.95	1.10			
	Ge(K)	4.93	3.84	0.36			
8-1	Pb(L)	12.35	3.37	0.78	$Zn_{0.6}Ge_{0.17}Pb_{0.08}S$	$4ZnS \cdot GeS_2 \cdot PbS$	1.96
	S(K)	31.61	53.82	0.92			
	Zn(K)	39.11	32.65	0.95			
	Ge(K)	11.88	8.94	0.59			
9	Pb(L)	17.37	4.57	0.93	$ZnGe_{0.06}Pb_{0.12}S$		0.48
	S(K)	25.15	45.74	0.20			
	Fe(K)	0.18	0.18	0.01			
	Zn(K)	51.27	45.73	0.28			
10-1	Ge(K)	3.34	2.69	0.07	$Zn_{0.5}Ge_{0.09}Pb_{0.13}S$		0.70
	Pb(L)	20.04	5.64	0.23			
	S(K)	32.64	58.26	1.91			
	Zn(K)	32.79	28.7	1.69			
10-2	Ge(K)	6.79	5.35	0.95	$Zn_{0.47}Ge_{0.08}Pb_{0.13}S$		0.69
	Pb(L)	27.76	7.66	2.26			
	S(K)	33.7	59.44	1.92			
	Zn(K)	32.26	27.91	1.68			
12-1	Ge(K)	6.62	5.15	0.94	$Zn_{0.85}Ge_{0.16}Pb_{0.16}S$		1.04
	Pb(L)	27.4	7.48	2.24			
	S(K)	24.32	46.1	0.89			
	Zn(K)	42.18	39.21	1.01			
12-2	Ge(K)	8.93	7.47	0.57	$Zn_{1.1}Ge_{0.13}Pb_{0.09}S$		1.44
	Pb(L)	24.55	7.20	1.06			
	S(K)	24.78	43.77	1.10			
	Zn(K)	54.13	46.88	1.42			
	Ge(K)	7.06	5.51	0.59			
	Pb(L)	14.01	3.82	1.14			

Note: K—spectral line K α ; L—spectral line L α .

detection of EDS (1–2 wt% Ge or Pb). An indirect method, comparing Ge/Pb ratios measured with LA-ICP-MS and EDS, was used to determine if the nanoparticles were the dominant hosts of Ge. The Ge/Pb ratios of sphalerite can reflect the amounts of nanoparticles. In Sp1a, Sp1b, and Sp2, the Ge and Pb contents measured with LA-ICP-MS and contents in nanoparticles measured with TEM-EDS were positively correlated ($R^2 = 0.85$) with a ratio of Ge/Pb ~ 1 (Fig. 14A). Furthermore, the LA-ICP-MS spot analyses in Sp1a, Sp1b, and Sp2 had Ge/Pb ratios similar to the TEM-EDS analyses on the nanoparticles (between 0.48 and 1.96, with a mean of 1.09; Fig. 14B). As the Pb is hardly incorporated into sphalerite (Cook et al., 2009; George et al., 2016), this result suggests that Ge and Pb mainly occur as Zn-Ge-Pb-S nanoparticles.

The extracted chemical composition from the LA-ICP-MS mapping of sphalerite, comprising >80,000 measurements, showed that Ge and Pb were positively correlated in Sp1b and Sp2 (Figs. 15A and 15D). The quantified compositional profiles (Figs. 15C and 15F) showed that Ge and Pb had similar distribution patterns in both Sp1b and Sp2. The calculated Ge/Pb ratios of Sp1b and Sp2 mainly ranged from 0.30 to 2.0 and from 0.36 to 2.0, respectively, which are consistent with the ratios in the Zn-Ge-Pb-S nanoparticles measured with TEM-EDS. These results likely suggest that the Zn-Ge-Pb-S nanoparticles are the predominant phase hosting Ge in the Banbianjie deposit.

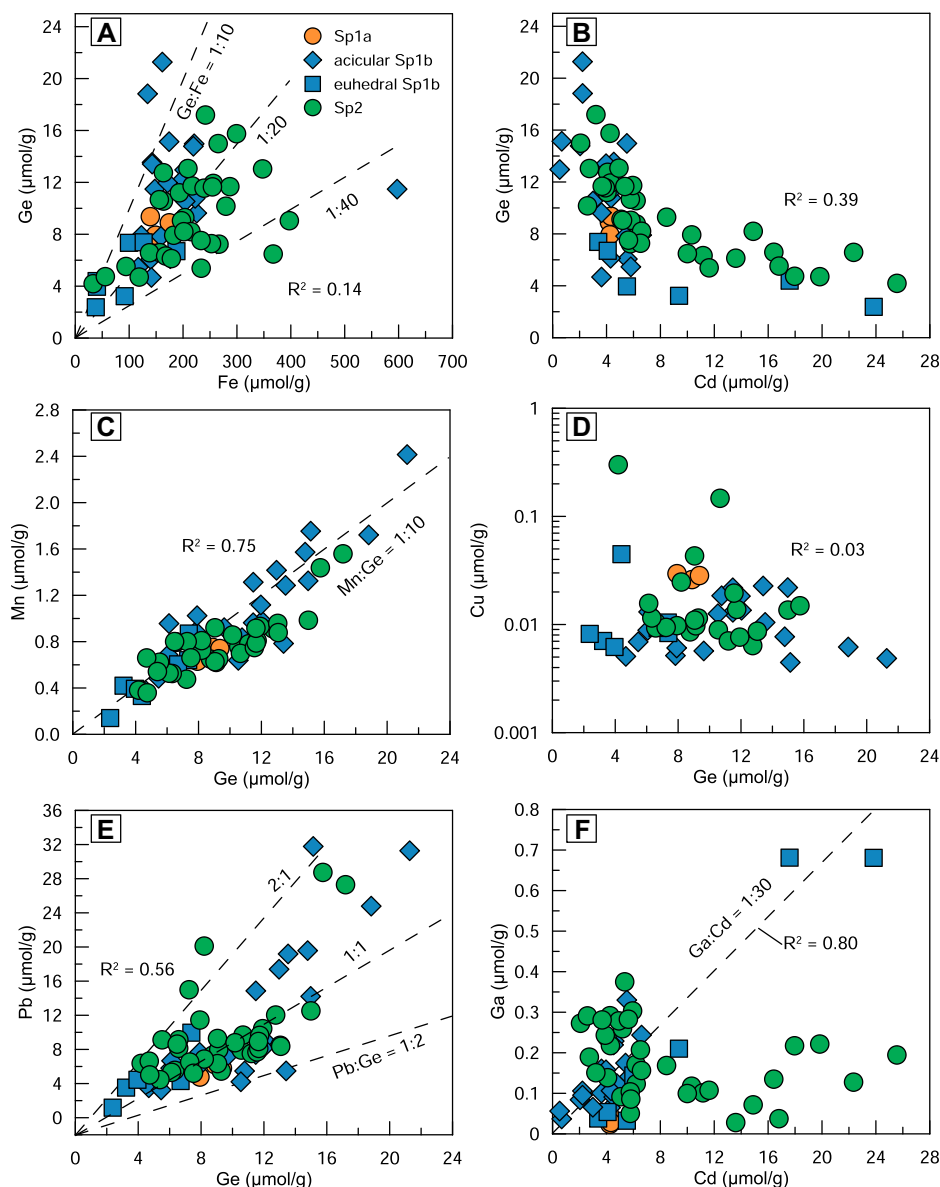


Figure 13. Binary plots for trace-element contents in sphalerite from the Banbianjie deposit. (A) Ge vs. Fe. (B) Ge vs. Cd. (C) Mn vs. Ge. (D) Cu vs. Ge. (E) Pb vs. Ge. (F) Ga vs. Cd. Sp1a—sphalerite 1a; Sp1b—sphalerite 1b; Sp2—sphalerite 2.

Formation of Ge Oscillatory Zoning in Sphalerite

Previous studies have suggested that oscillatory chemical zoning observed in minerals, including sphalerite and pyrite, may reflect self-organized nonequilibrium processes during crystal growth (Fowler and L'Heureux, 1996; Di Benedetto et al., 2005; Wu et al., 2019; Fougereuse et al., 2023). In the Banbianjie deposit, the colloform texture may suggest a rapid growth rate (Abraitis et al., 2004; Franchini et al., 2015), which may have been faster than the diffusion rate and allowed self-organization processes

(Fowler and L'Heureux, 1996; Phelps et al., 2020). The Fe (Mn) and Cd patterns are antithetic in sphalerite from Banbianjie (Figs. 9, 10, 11, and 15), suggesting that Fe (Mn) may compete with Cd to replace Zn. Both Fe^{2+} (0.63 Å) and Mn^{2+} (0.66 Å) have ionic radii similar to Zn^{2+} (0.60 Å), whereas Cd^{2+} (0.78 Å) is slightly larger than Zn^{2+} . This difference in ionic radius could explain the initial zoning in sphalerite, which is usually Fe- and Mn-rich and Cd-poor (Figs. 9, 10, and 16A). When the fluid becomes depleted in Fe (Mn), Cd starts to be incorporated into sphalerite, following a typical feedback reaction pathway (Figs. 16A and 16B). The feedback pro-

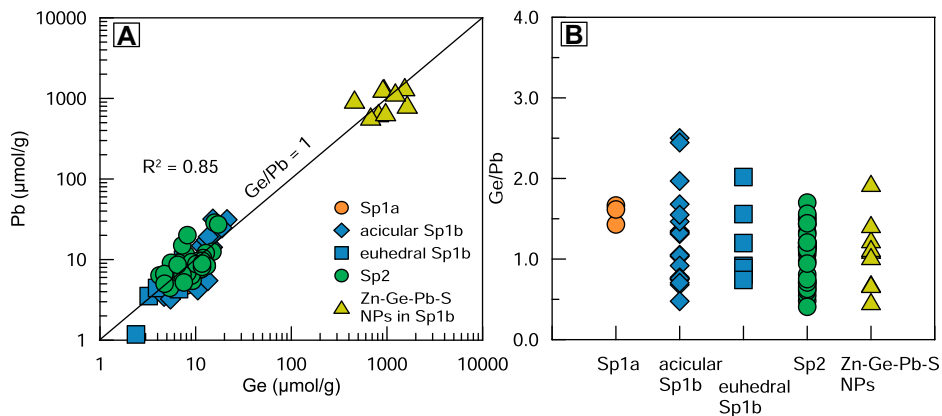
cesses represent the self-organization that controls the fluid composition at the boundary layer and, consequently, the formation of oscillatory zoning textures (Di Benedetto et al., 2005). Similarly, the Ga^{3+} (0.47 Å) and In^{3+} (0.62 Å) ions may also compete with each other to combine with monovalent cations (e.g., Cu^+ and Ag^+) and to substitute for Zn^{2+} (0.60 Å). In Sp1b, Ga and In are anticorrelated (Fig. 9), and the initial zoning of Sp1b is In-rich (Fig. 9), which is consistent with the fact that the size of In^{3+} is very similar to that of Zn^{2+} , compared to the ionic radius of Ga^{3+} . In Sp2 (Fig. 10), In contents were very low (0.01–0.87 ppm), probably due to the primary fluid composition preventing competition and self-organization processes between Ga and In.

The Ge oscillatory zoning showed a correlation with Fe and Mn contents, suggesting that the Zn-Ge-Pb-S nanoparticles may also be a consequence of self-organization. In our model, self-organization controlled the element concentrations in the boundary layer. Abundant Zn^{2+} , Fe^{2+} , Mn^{2+} , Ge^{4+} , and Pb^{2+} in the boundary layer may have reacted with HS^- , triggering the precipitation of Zn-Ge-Pb-S nanoparticles and acicular sphalerite. Because the growth rate was faster than the diffusion rate, cations such as Zn, Ge, Fe, Mn, and Pb were likely depleted at the boundary layer (Phelps et al., 2020), which resulted in a decrease of the abundance of Zn-Ge-Pb-S nanoparticles and slower growth rates of euhedral sphalerite (Figs. 16A and 16B). This process was cyclical, and before the next precipitation event, the concentrations of Zn, Ge, Fe, Mn, and Pb in the boundary layer would have increased to form a new acicular sphalerite layer rich in nanoparticles (Figs. 16A and 16B).

In classical carbonate-hosted Pb-Zn deposits, Ge is commonly described as occurring in the sphalerite crystal lattice coupled with other metal ions (e.g., Cu^+ , Ag^+ , Mn^{2+} , Pb^{2+} ; Belisont et al., 2016; Luo et al., 2022; Liu et al., 2023; Torró et al., 2023). In our study, we showed that Ge mostly occurs in Zn-Ge-Pb-S nanoparticles within sphalerite. A sphalerite poor in Cu and fast crystal growth may be prerequisites for the formation of nanoparticles. Other particular physico-chemical conditions (e.g., pH, redox, fluid compositions) may finally result in the precipitation of the nanoparticles. Additional experiments are needed to better constrain their mode of occurrence and distribution, considering that they represent low-temperature hydrothermal systems in which Ge is highly enriched.

CONCLUSIONS

Sphalerite from the Banbianjie deposit showed high contents of Ge (up to 1553 ppm).



The distribution of Ge in colloform sphalerite showed oscillatory zoning with higher Ge enrichment in darker sphalerite, positive correlations between Ge and Fe, Pb, and Mn, and a negative correlation with Cd. We documented the occurrence of Zn-Ge-Pb-S nanoparticles. Based on TEM and LA-ICP-MS analyses, these nanoparticles may be the dominant host for Ge due to the similar Ge/Pb ratios measured in both the Zn-Ge-Pb-S nanoparticles and the host sphalerite. The oscillatory chemical zoning (including Ge) in sphalerite may be related to self-organization processes, which resulted in the fluctuations of various trace-element concentrations in the boundary layer. The element concentrations in the boundary layer may further have influenced the growth rate and texture of the sphalerite. This study

Figure 14. (A) Binary plot for Pb vs. Ge. (B) Ge/Pb ratios for different types of sphalerite and Zn-Ge-Pb-S nanoparticles (NPs). Sp1a—sphalerite 1a; Sp1b—sphalerite 1b; Sp2—sphalerite 2.

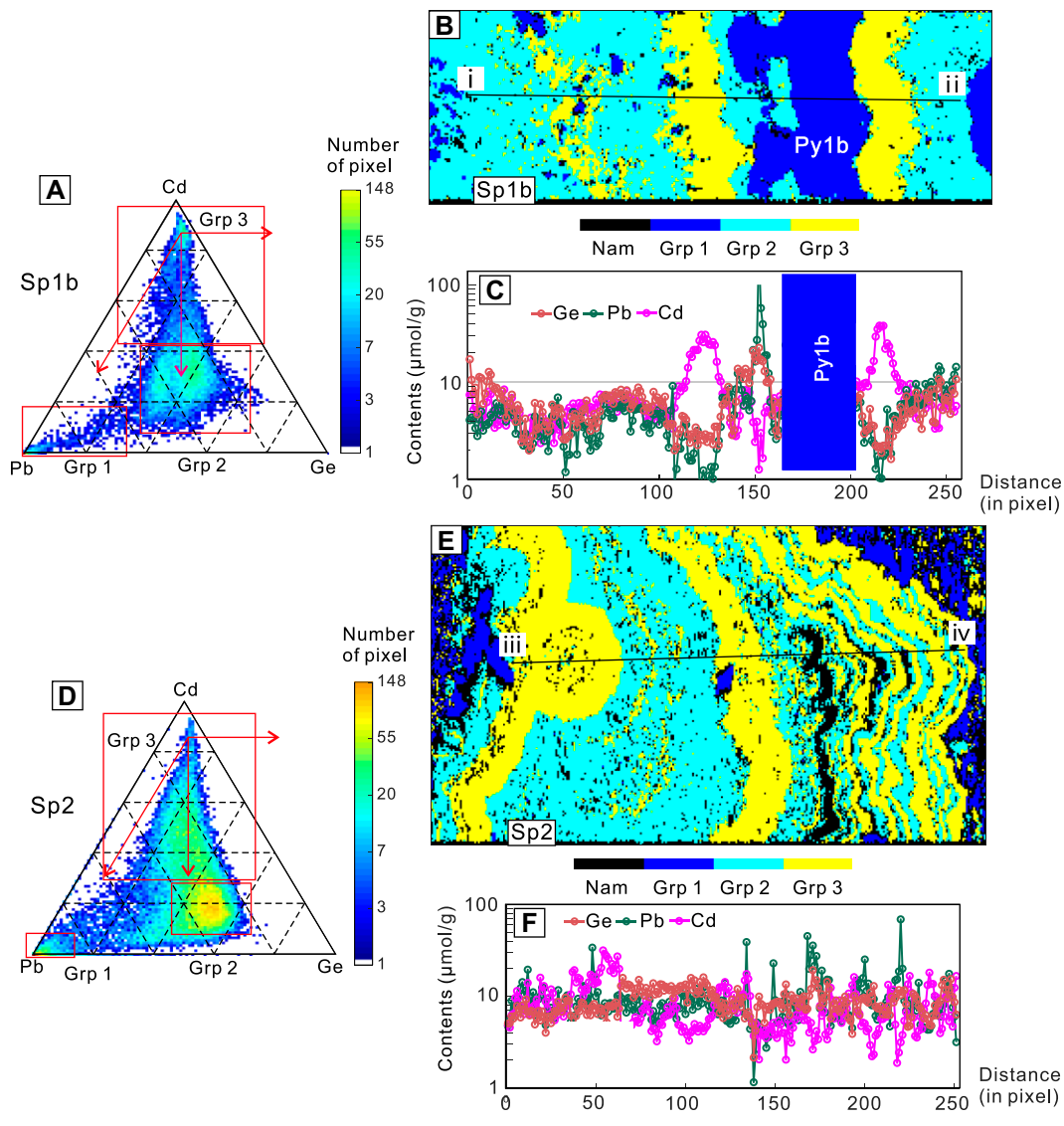


Figure 15. Comparison of Cd, Ge, and Pb contents (μmol/g) in sphalerite. (A) Ternary diagram showing three compositional groups (Grp). Grp1 with high Pb contents represents pyrite. Because we used $^{66}\text{Zn} = 650,000$ ppm as an internal standard, pyrite has overestimated Pb contents relative to sphalerite. Grp 2 has higher Ge and Pb and lower Cd contents than Grp3. Nam represents the pixel that is not included in the above three groups. (B) Pixel identification map with the three groups discriminated in panel A. (C) Laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) data (μmol/g) values across the line i–ii indicated in panel B. (D) Ternary diagram showing three groups. Grp1, with high Pb contents, represents galena and dolomite. Grp 2 has higher Ge and Pb and lower Cd contents than Grp3. Larger Ge contents in Sp2 are noted compared to Sp1b. (E) Pixel identification map with the three groups discriminated in panel D. (F) LA-ICP-MS data (μmol/g) values across the line iii–iv indicated in panel E. Abbreviations: Py1b—pyrite 1b; Sp1b—sphalerite 1b; Sp2—sphalerite 2.

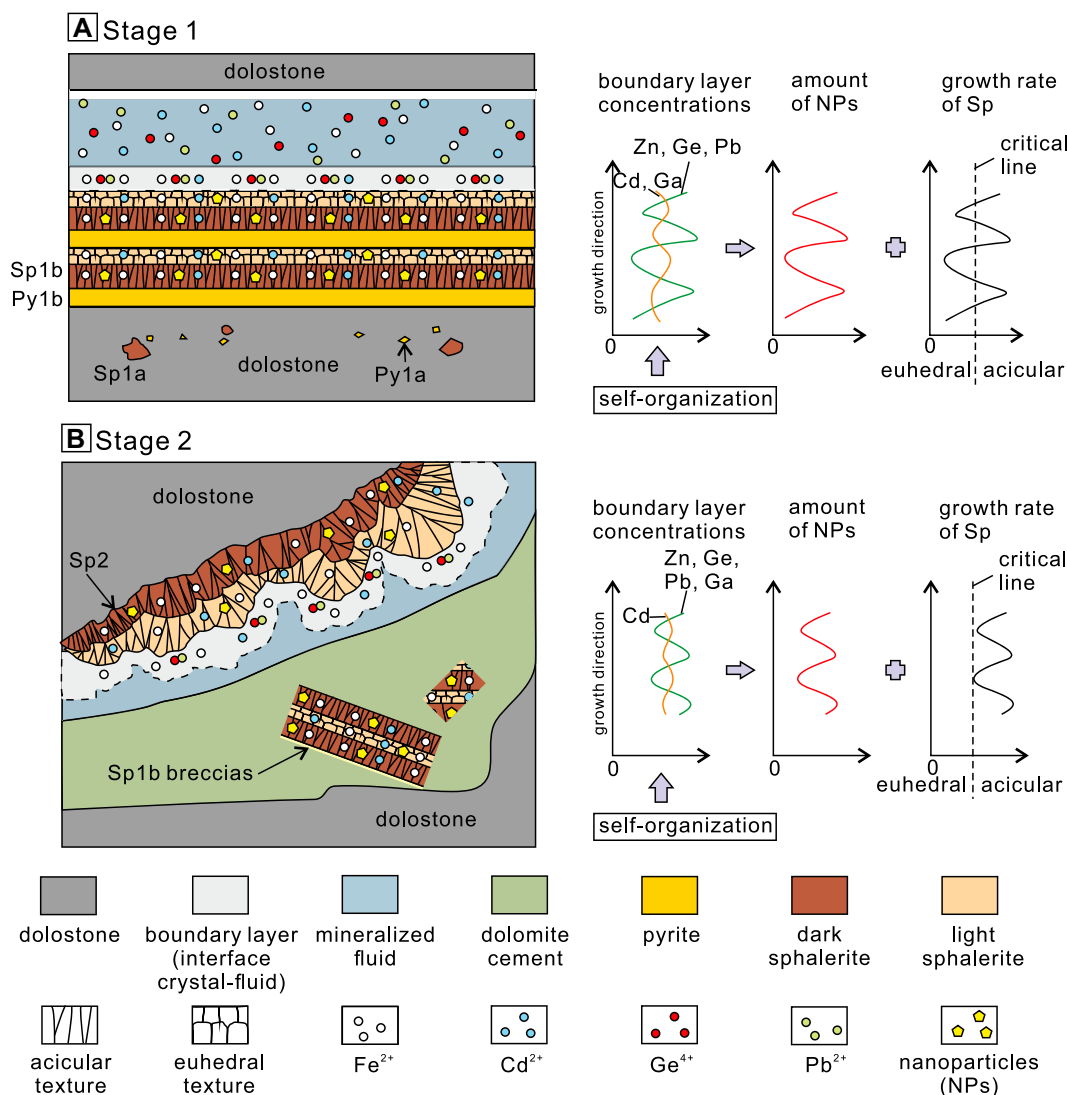


Figure 16. Schematic model of Ge enrichment in sphalerite from the Banbianjie deposit. On the right, the diagrams illustrate the relative changes in Zn-Ge-Pb and Cd-Ga concentrations, in the abundance of nanoparticles (NPs), and in the sphalerite (Sp) growth rate. (A) In stage 1, acicular and euhedral grains of Sp1b show oscillatory zoning. (B) In stage 2, acicular crystals of Sp2 exhibit only oscillatory zoning. Brecciated clasts of Sp1 can be seen on the right of this sketch. Py—pyrite; Sp1b—sphalerite 1b; Sp1a—sphalerite 1a; Sp2—sphalerite 2.

showed the ability of nanoscale studies to unravel the incorporation mechanisms of critical elements in sphalerite, a key component for critical raw material exploration and ore processing.

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